

**MASS**

A measure of the amount of matter in an object.

MATERIAL

A substance out of which things can be made. Every object is made of a material—natural or human-made.

MATTER

The material which everything around us is made from. Anything that has mass and occupies space is matter. Matter includes solids, liquids, and gases, and both living and nonliving things.

METAL

A type of element with specific properties. Metals are usually strong and shiny and are good conductors of electricity and heat. Over three-quarters of all elements are classified as metals.

MICROBE

A tiny organism that can only be seen with the aid of a microscope. Also known as a microorganism.

MICROORGANISM

See microbe

MINERAL

Any naturally occurring substance made up of more than one element that can be found in the ground. Rocks are made of minerals. All minerals are crystals.

MOLD

A type of fungus. It appears on rotten food as a fluffy growth. Most molds feed on decaying plants and animals.

MOLECULE

A group of two or more atoms joined by strong bonds.

MOMENTUM

The measure of an object's motion. Momentum is calculated by multiplying the object's mass by its speed.

MOTOR

A device that uses electricity and magnetism to produce movement.

MUSCLE

Organs in the body made up of tiny fibers. Muscles pull on bones to make the body move.

NERVE

A bundle of nerve fibers and blood vessels that transmits electrical signals to and from the brain or the spinal cord.

NEURON

A nerve cell. Each neuron is connected to many other neurons, creating a network for sending and receiving signals.

NEUTRON

A particle in the nucleus of an atom that has no electric charge. It is located in the nuclei of all atoms except hydrogen.

NUCLEAR FISSION

A process in which large nuclei of atoms break into smaller ones, releasing large amounts of energy.

NUCLEAR FUSION

A process in which small atomic nuclei, such as those of hydrogen atoms, join together to make larger ones, releasing large amounts of energy.

NUCLEUS

The central core of something. An atomic nucleus contains protons and neutrons, while a cell's nucleus contains DNA.

NUTRIENT

A substance that animals and plants take in that is essential for life and growth. Nutrients are useful as a source of energy or as raw material.

ORGAN

A major structure in an organism that has a specific function. Examples include the stomach, the brain, and even the skin.

ORGANISM

Any living thing, such a plant, animal, or fungus.

OXYGEN

A colorless, odorless gas. It is required for a substance to burn.

PARTICLE

A tiny bit of matter. This can refer to atoms and molecules or to subatomic particles such as electrons, protons, and neutrons.

PATHOGEN

A microbe, such as a virus, that causes disease.

PH

A measure of the concentration of charged atoms of hydrogen that determines how acidic or alkaline a substance is. The lower the pH, the more acidic the substance.

PHOTOSYNTHESIS

The process by which plants use the Sun's energy to make food from water and carbon dioxide. This process produces oxygen.

PHYSICS

The study of forces, energy, and matter.

PIGMENT

A chemical that colors an object.

PLASMA

A high-energy gas made of charged particles. Also the liquid part of blood.

POLYMER

A long, chainlike molecule made up of smaller molecules connected together. Polymers can be found in nature, such as DNA, or be produced artificially, such as plastic.

PRESSURE

The force exerted by something pressing or squeezing an area. The same force can produce high pressure or low pressure, depending on the area it acts on.

PROTEIN

A type of complex chemical found in all living things. Proteins are the building blocks of cells. These are made up of simple units called amino acids. There are about 20 natural amino acids, and a protein has hundreds of these units connected in a specific order. Organisms need proteins for growth and repair.

PROTON

A particle with a positive electric charge in the nucleus of an atom.

RADIATION

Waves of energy that travel through space. Radiation includes visible light, heat, X-rays, and radio waves.

RADIOACTIVE

A material that is unstable because the nuclei of its atoms easily break down.

RADIOACTIVITY

The breakdown of atomic nuclei, causing radiation to be released.